Example contd

Let $\mathfrak{g} = \mathfrak{sl}_3$ and $\lambda = (2,1)$.

Then $t_+ = E_{13}$, $t_- = E_{31}$, $t_0 = E_{11} - E_{33}$.

$\xi = \frac{1}{3}(E_{11} + E_{22} - 2E_{33})$

$\mathfrak{m} = \langle \{E_{13}, E_{23}\} \rangle = \mathfrak{g}_+$

$\chi(E_{13}) = 1, \quad \chi(E_{23}) = 0$

$\Rightarrow \mathfrak{g}_- = \langle \{E_{31}, E_{32}\} \rangle$.

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ -1 & -1 & 0 \end{bmatrix}$$

grading w.r.t. adg.

$$Q_0 = E_{13} \otimes \xi_{13}^* + E_{23} \otimes \xi_{23}^*, \quad Q_1 = -1 \otimes \xi_{13}^*$$

$$\begin{cases} d_0(E_{ij} \otimes 1) = [E_{13}, E_{ij}] \otimes \xi_{13}^* + [E_{23}, E_{ij}] \otimes \xi_{23}^*, \\ d_0(1 \otimes \xi_{13}^*) = d_0(1 \otimes \xi_{23}^*) = 0, \\ d_0(1 \otimes \xi_{13}) = E_{13} \otimes \xi_{13}^* \xi_{13} + E_{23} \otimes \xi_{23}^* \xi_{13} + E_{13} \otimes \xi_{13} \xi_{13}^* + E_{23} \otimes \xi_{13} \xi_{23}^* \\ \qquad\qquad = E_{13} \otimes 1 \\ d_0(1 \otimes \xi_{23}) = E_{23} \otimes 1 \end{cases}$$

$$\begin{cases} d_1(E_{ij} \otimes 1) = d_1(1 \otimes \xi_{13}^*) = d_1(1 \otimes \xi_{23}^*) = 0, \\ d_1(1 \otimes \xi_{13}) = -1, \quad d_1(1 \otimes \xi_{23}) = 0 \end{cases}$$

A new set of generators

Let $\psi_\alpha = 1 \otimes \xi_\alpha$, $\psi_\alpha^* = 1 \otimes \xi_\alpha^*$, and

$$J_a = (x_a \otimes 1) + \sum_{\alpha,\beta} f_{a\alpha}^{\beta} \psi_\beta \psi_\alpha^* .$$

Lemma

We have :

$$d_0(J_a) = \sum_{\alpha,\beta} f_{\alpha a}^{\beta} J_{\beta} \psi_{\alpha}^{*} , \quad d_1(J_a) = -\sum_{\alpha,\beta} f_{a\alpha}^{\beta} \chi(x_\beta) \psi_{\alpha}^{*} ,$$

$$d_0(\psi_\alpha^*) = -\frac{1}{2} \sum_{\beta,\gamma} f_{\beta\gamma}^{\alpha} \psi_\beta^* \psi_\gamma^* , \quad d_1(\psi_\alpha^*) = 0 ,$$

$$d_0(\psi_\alpha) = J_\alpha , \quad d_1(\psi_\alpha) = -\chi(x_\alpha) .$$

∎

Example Contd

Choose a basis of Cartan subalgebra to be $\{E_{11}-2E_{22}+E_{33}, E_{11}-E_{33}\}$.

$$J_0 = (E_{11}-2E_{22}+E_{33}) \otimes 1 - 3\,\psi_{23}\psi_{23}^*$$

$$J_{01} = (E_{11}-E_{33}) \otimes 1 + 2\,\psi_{13}\psi_{13}^* + \psi_{23}\psi_{23}^*$$

$$J_{21} = E_{21} \otimes 1 + \psi_{23}\psi_{13}^* , \quad J_{32} = E_{32} \otimes 1 , \quad J_{31} = E_{31} \otimes 1$$

$$J_{12} = E_{12} \otimes 1 + \psi_{13}\psi_{23}^*$$

$$d_0(J_0) = 3\,E_{23} \otimes \xi_{23}^* - 3\,E_{23} \otimes \xi_{23}^* = 0 ,$$

$$d_0(J_{01}) = -E_{13} \otimes \xi_{13}^* - E_{23} \otimes \xi_{23}^* ,$$

$$d_0(J_{21}) = -E_{23} \otimes \xi_{13}^* + E_{23} \otimes \xi_{13}^* = 0 ,$$

$$d_0(J_{32}) = E_{12} \otimes \xi_{13}^* + (E_{22}-E_{33}) \otimes \xi_{23}^* ,$$

$$d_0(J_{31}) = (E_{11}-E_{33}) \otimes \xi_{13}^* + E_{21} \otimes \xi_{23}^* ,$$

$$d_0(J_{12}) = 0 .$$

$$d_1(J_0) = d_1(J_{21}) = d_1(J_{32}) = d_1(J_{31}) = 0$$

$$d_1(J_{01}) = -2\,\psi_{13}^* , \quad d_1(J_{12}) = -\psi_{23}^*$$

Example Contd.

$$\mathcal{A}^{\mathrm{gf}} = \left\langle \{ E_{11} - 2E_{22} + E_{33},\ E_{21},\ E_{32},\ E_{31} \} \right\rangle$$

$\updownarrow$

4 generators of $H^*(S^2, d)$

$\{ \hat{J}_0,\ \hat{J}_{21},\ \hat{J}_{32},\ \hat{J}_{31} \}$

$$J_0 = (E_{11} - 2E_{22} + E_{33}) \otimes 1 - 3\, \psi_{23}\, \psi_{23}^*$$
$$J_{01} = (E_{11} - E_{33}) \otimes 1 + 2\, \psi_{13}\, \psi_{13}^* + \psi_{23}\, \psi_{23}^*$$
$$J_{21} = E_{21} \otimes 1 + \psi_{23}\, \psi_{13}^* \ , \quad J_{32} = E_{32} \otimes 1 \ , \quad J_{31} = E_{31} \otimes 1$$
$$J_{12} = E_{12} \otimes 1 + \psi_{13}\, \psi_{23}^*$$

$$d_0(J_0) = 3\, E_{23} \otimes \xi_{23}^* - 3\, E_{23} \otimes \xi_{23}^* = 0 \ ,$$
$$d_0(J_{01}) = -E_{13} \otimes \xi_{13}^* - E_{23} \otimes \xi_{23}^* \ ,$$
$$d_0(J_{21}) = -E_{23} \otimes \xi_{13}^* + E_{23} \otimes \xi_{13}^* = 0 \ ,$$
$$d_0(J_{32}) = E_{12} \otimes \xi_{13}^* + (E_{22} - E_{33}) \otimes \xi_{23}^* \ ,$$
$$d_0(J_{31}) = (E_{11} - E_{33}) \otimes \xi_{13}^* + E_{21} \otimes \xi_{23}^* \ ,$$
$$d_0(J_{12}) = 0 \ .$$

$$d_1(J_0) = d_1(J_{21}) = d_1(J_{32}) = d_1(J_{31}) = 0$$
$$d_1(J_{01}) = -2\, \psi_{13}^* \ , \quad d_1(J_{12}) = -\psi_{23}^*$$

Tic-tac-toe :

$$\hat{J}_0 = J_0 = J_{11} - 2J_{22} + J_{33} \ , \quad \hat{J}_{21} = J_{21}$$

$$d_0(J_{32}) = E_{12} \otimes \xi_{13}^* + (E_{22} - E_{33}) \otimes \xi_{23}^*$$
$$= d_1\left(-E_{12} \otimes \xi_{13}\, \xi_{13}^* - (E_{22} - E_{33}) \otimes \xi_{13}\, \xi_{23}^* + \cdots \right)$$
$$= d_1\left(\tfrac{-1}{2} J_{12} J_{01} + \tfrac{1}{2} J_0 J_{12} + \tfrac{1}{2} J_{12} \right)$$

$$\Rightarrow \hat{J}_{32} = J_{32} + \tfrac{1}{2} J_{12} J_{01} - \tfrac{1}{2} J_0 J_{12} - \tfrac{1}{2} J_{12}$$

$$d_0(J_{31}) = (E_{11} - E_{33}) \otimes \xi_{13}^* + E_{21} \otimes \xi_{23}^*$$
$$= d_1\left(-(E_{11} - E_{33}) \otimes \xi_{13}\, \xi_{13}^* - E_{21} \otimes \xi_{13}\, \xi_{23}^* \right)$$
$$= d_1\left(\tfrac{-1}{4} J_{01}^2 - J_{12} J_{21} + J_{01} \right)$$

$$\Rightarrow \hat{J}_{31} = J_{31} + \tfrac{1}{4} J_{01}^2 + J_{12} J_{21} - J_{01}$$

Relations

Note that $[J_\alpha, J_\beta] = \sum_\gamma f_{\alpha\beta}^{\gamma} J_\gamma$

$$U(\mathcal{T}, f) := H^0(\Omega, d) = U(\mathcal{T}^f) / \text{relations}$$

$$= \langle \hat{J}_0, \hat{J}_{21}, \hat{J}_{32}, \hat{J}_{31} \rangle / (\dots)$$

$$\hat{J}_{31} = J_{31} + \frac{1}{4} J_{01} J_{01} + J_{12} J_{21} - J_{01}$$

$$\hat{J}_{32} = J_{32} + \frac{1}{2} J_{12} J_{01} - \frac{1}{2} J_0 J_{12} - \frac{1}{2} J_{12}$$

$$\hat{J}_0 = J_0, \quad \hat{J}_{21} = J_{21}$$

$$C = -\frac{4}{3} \hat{J}_{31} - \frac{1}{9} \hat{J}_0 \hat{J}_0 - 1$$

$$H = -\frac{2}{3} \hat{J}_0 - 1$$

$$E = \frac{4}{3} \hat{J}_{21}, \quad F = \hat{J}_{32}$$

$J_{11} - J_{22} = \frac{1}{2}(J_0 + J_{01})$

$[J_0, J_{12}] = 3 J_{12}$, $[J_{01}, J_{12}] = J_{12}$

$[J_0, J_{21}] = -3 J_{21}$, $[J_{01}, J_{21}] = -J_{21}$

$[J_0, J_{32}] = 3 J_{32}$, $[J_{01}, J_{32}] = -J_{32}$

$[J_0, J_{31}] = 0$, $[J_{01}, J_{31}] = -2 J_{31}$

$[J_0, J_{12} J_{21}] = 0$, $[J_{01}, J_{12} J_{21}] = 0$

(1) $[\hat{J}_0, \hat{J}_{21}] = -3 \hat{J}_{21}$, $[\hat{J}_0, \hat{J}_{32}] = 3 \hat{J}_{32}$, $[\hat{J}_0, \hat{J}_{31}] = 0$

$[\hat{J}_{31}, \hat{J}_{21}] = \frac{3}{4} \hat{J}_{21} + \frac{1}{2} \hat{J}_0 \hat{J}_{21}$, $[\hat{J}_0^2, \hat{J}_{21}] = -9 \hat{J}_{21} - 6 \hat{J}_0 \hat{J}_{21}$

$[\hat{J}_{31}, \hat{J}_{32}] = \frac{3}{4} \hat{J}_{32} - \frac{1}{2} J_0 \hat{J}_{32}$, $[\hat{J}_0^2, \hat{J}_{32}] = -9 \hat{J}_{32} + 6 \hat{J}_0 \hat{J}_{32}$

$$\left[\hat{J}_{31} + \frac{1}{12} \hat{J}_0^2, \hat{J}_{21}\right] = \left[\hat{J}_{31} + \frac{1}{12} \hat{J}_0^2, \hat{J}_{32}\right] = 0 \quad (2)$$

$$[\hat{J}_{21}, \hat{J}_{32}] = -\hat{J}_{31} + \frac{1}{4} \hat{J}_0^2 + \hat{J}_0 \quad (3)$$

Then, $[H, E] = 2E$, $[H, F] = -2F$, $[E, F] = H^2 + C$,

$[C, E] = [C, F] = [C, H] = 0$.

The Miura Transformation

Theorem

Let $W(\phi) \in H^0(\Omega; d)$ be the representative of $\phi \in U(\mathfrak{f})$.

Denote by $X^{(0,0)}$ the component of $X \in H^0(\Omega; d)$ having bidegree $(0,0)$.

Then the map $W(\phi) \longmapsto W(\phi)^{(0,0)}$ is an isomorphism of algebras.

or

$$M : H^0(\Omega; d) \longrightarrow H^0(\Omega; d)^{(0,0)}$$



(*) This map gives a faithful realization of $U(\mathfrak{f}, f)$ in $U(\mathfrak{f}_0)$, which is a direct sum of simple Lie algebras (upto $U(1)$ terms).

$$U(\mathfrak{f}, f) \xrightarrow[\text{map}]{\text{Miura}} U(\mathfrak{f}_0) \longrightarrow V$$

(*) For principal sl_2 embedding, $U(\mathfrak{f}_0)$ is abelian.

$\Rightarrow U(\mathfrak{f}, f)$ is abelian. In this case, $U(\mathfrak{f}, f) \cong Z(U(\mathfrak{g}))$ (Kostant)

Example contd.

$$\hat{J}_{31} = J_{31} + \frac{1}{4} J_{01} J_{01} + J_{12} J_{21} - J_{01}$$
$$\hat{J}_{32} = J_{32} + \frac{1}{2} J_{12} J_{01} - \frac{1}{2} J_0 J_{12} - \frac{1}{2} J_{12}$$
$$\hat{J}_0 = J_0 \,, \quad \hat{J}_{21} = J_{21}$$

$$C = -\frac{4}{3} \hat{J}_{31} - \frac{1}{9} \hat{J}_0 \hat{J}_0 - 1$$
$$H = -\frac{2}{3} \hat{J}_0 - 1$$
$$E = \frac{4}{3} \hat{J}_{21} \,, \quad F = \hat{J}_{32}$$

$$M(\hat{J}_0) = \hat{J}_0 \,, \quad M(\hat{J}_{21}) = \hat{J}_{21} \,,$$

$$M(\hat{J}_{31}) = \frac{1}{4} J_{01}^2 + J_{12} J_{21} - J_{01} \,, \quad M(\hat{J}_{32}) = \frac{1}{2} J_{12} J_{01} - \frac{1}{2} J_0 J_{12} - \frac{1}{2} J_{12}$$

$$\Rightarrow M(H) = -\frac{2}{3} \hat{J}_0 - 1 \,, \quad M(E) = \frac{4}{3} \hat{J}_{21} \,, \quad M(F) = \frac{1}{2} J_{12} J_{01} - \frac{1}{2} J_0 J_{12} - \frac{1}{2} J_{12}$$

$$M(C) = -\frac{4}{3}\left(\frac{1}{4} J_{01}^2 + J_{12} J_{21} - J_{01}\right) - \frac{1}{9} J_0^2 - 1 = \frac{1}{2} J_{01} J_{12} - \frac{1}{2} J_0 J_{12} - J_{12} \,,$$

Let $s = \frac{1}{4}\left(\hat{J}_0 - 3\hat{J}_{01}\right)$, $h = \frac{1}{2}\left(\hat{J}_0 + \hat{J}_{01}\right)$, $e = \hat{J}_{12}$, $f = \hat{J}_{21}$

$\Rightarrow [e, f] = h$, $[h, e] = 2e$, $[h, f] = -2f$, and s commutes with all

$\left(\mathfrak{g}_0 = \mathfrak{sl}_2 \oplus \mathfrak{u}(1)\right)$

$$M(E) = \frac{4}{3} f \,, \quad M(H) = -h - \frac{2}{3} s - 1 \,, \quad M(f) = \left(-\frac{1}{2} h - s - 1\right) e$$

$$M(C) = -\frac{4}{3}\left(h^2 + \frac{1}{2} ef + \frac{1}{2} fe\right) + h^2 - \frac{4}{9} s^2 - \frac{4}{3} s - 1$$

$$-\frac{4}{3}s + \frac{4}{3}\left(\frac{1}{4}T_0^2\right) + \frac{4}{3}\left(\frac{1}{2}T_0 T_{01}\right) - \frac{1}{9}T_0^2 - 1$$

$$= -\frac{4}{3}s + \frac{2}{9}T_0^2 + \frac{2}{3}T_0 T_{01} - 1$$

$$T_0 = \frac{1}{4}(4s+6h), \quad T_{01} = \frac{1}{4}(2h-4s)$$

$$\frac{2}{3}\cdot\frac{1}{4}(4s+6h)\left[\frac{1}{12}(4s+6h) + \frac{1}{4}(2h-4s)\right]$$

$$\frac{1}{6}(4s+6h)\left[\frac{1}{3}s + \frac{1}{2}h + \frac{1}{2}h - s\right]$$

$$\frac{1}{3}(2s+3h)\left(h - \frac{2}{3}s\right)$$

$$\frac{1}{9}(2s+3h)(3h-2s) = \frac{1}{9}(9h^2 - 4s^2)$$

$$= -\frac{4}{9}s^2 + h^2$$